Zakarya Boudraf

Junior Machine Learning Engineer

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PROFILE

Machine-learning engineer with an M.Sc. in Computer Science (Internet of Things curriculum, final grade 108/110, taught in English) earned in July 2026 at the University of Salerno, and a prior M.Sc. in Data Engineering and Web Technologies earned Magna Cum Laude (ranked 2nd of 72). Research experience spans audio-visual speech recognition with parameter-efficient fine-tuning of transformer backbones, computer vision, and biomedical signal classification, with hands-on model deployment from microcontrollers to the web. Available immediately for entry-level machine learning roles in Italy: on-site in the Salerno area, or hybrid and remote elsewhere.

EDUCATION

M.Sc. Computer Science - Internet of Things curriculum (Laurea Magistrale in Informatica, LM-18)

University of Salerno, Italy [10/2023 - 07/2026] EQF 7 · taught in English

Final grade: 108/110. Thesis: Efficient Visual Speech Recognition using Parameter-Efficient Fine-Tuning of Frozen Self-Supervised Backbones (AV-HuBERT). Advisor: Prof. Fabio Narducci.

M.Sc. Data Engineering and Web Technologies

Ferhat Abbas Setif 1 University, Algeria [09/2021 - 07/2023] EQF 7

Magna Cum Laude - ranked 2nd of 72 students (top 3%), average grade 14.49/20. Thesis: self-supervised learning for epileptic seizure detection.

B.Sc. Computing Systems

Ferhat Abbas Setif 1 University, Algeria [09/2018 - 09/2021] EQF 6

Magna Cum Laude - ranked 2nd of 334 students (top 1%), average grade 15.65/20.

RESEARCH AND MACHINE LEARNING PROJECTS

Efficient visual speech recognition via PEFT (M.Sc. thesis, University of Salerno) [12/2025 - 07/2026]

- Adapted a frozen AV-HuBERT transformer backbone to visual-only lip reading using LoRA and adapter modules (PyTorch, fairseq), evaluated across three languages of the multilingual MuAViC benchmark.
- Matched full fine-tuning at 0.77 character error rate while training 33M parameters instead of 344M, roughly ten times fewer.

Real-time IoT intrusion detection on STM32 · code [2026]

- Built a network intrusion-detection system with a neural network trained and running on an STM32 Nucleo-F401RE (ARM Cortex-M4, C/C++, TinyML), using the KDD Cup 1999 dataset.
- Reached 22 microsecond inference in roughly 2 KB of RAM, with zero false positives across 100 real-world packets.

Deep reinforcement learning for emergency traffic control · code [2026]

- Trained PPO agents in Unity ML-Agents to coordinate two consecutive intersections, cutting ambulance stops from 85% under a fixed timer to under 5% while reducing average waiting time by about 35%.

Smart monitoring system for hospital wards (team project) [2025]

- Designed an IoT ward-monitoring system of smart beds and a nurses' station dashboard, then ran a usability pilot with 12 clinical participants (nurses, physicians, IT staff and an administrator).

Data-centric ML for industrial predictive maintenance [2025]

- Systematic literature review following the Kitchenham protocol, screening 974 papers down to 23 primary studies; presented at the International School of IoT.

Cleaning robot built with test-driven development · code [2024]

- Implemented six user stories for an embedded cleaning robot in Python, writing a failing test before every feature and mocking the GPIO layer so behaviour could be verified without the hardware attached.

AI-generated art detection (team project) · live demo on Hugging Face Spaces [2024]

- Trained and compared CNN, VGGNet, ResNet and DenseNet on the public AI-ArtBench dataset (TensorFlow/Keras), reaching 94% accuracy.
- Deployed the ResNet checkpoint as a public interactive demo (Gradio) with real-time predictions and confidence scores; presented at a University of Salerno research seminar (07/2024).

Self-supervised learning for epileptic seizure detection (M.Sc. thesis, UFAS) [2022 - 2023]

- Applied self-supervised pretraining to EEG signal classification (TensorFlow/Keras), achieving 99.08% accuracy, 83.33% sensitivity and a 0.06 false-positive rate.

Additional projects: IoT fire-detection system with MQTT monitoring and automated fan control (Arduino, 2024); commercial website for an architecture studio (2019).

WORK EXPERIENCE

Web developer

Sacom Integrale Pub, Setif, Algeria [06/2022 - 09/2022]

- Delivered client websites end-to-end, independently managing complete projects.
- Gathered requirements and negotiated project details directly with customers.

TECHNICAL SKILLS

Machine learning and deep learning: TensorFlow · Keras · PyTorch · fairseq · CNNs · transformers · self-supervised learning · PEFT (LoRA, adapters) · reinforcement learning

Programming and data: Python · NumPy · Pandas · SQL · C/C++ · Java · JavaScript

Practices and tools: Git and GitHub · GitHub Actions · test-driven development · Gradio · Hugging Face Spaces

Embedded and IoT: STM32 · Arduino · Raspberry Pi · Unity ML-Agents · MQTT · sensor integration

Web: HTML/CSS · React · WordPress

LANGUAGES

Arabic (native) | English | Italian | French